STAT 429: Introduction to Time Series
Week 7 Lecture Notes

University of Illinois Urbana-Champaign

Fall 2026

1 Variance Models

Our next topic involves Variance Models (especially in the context of spectral analysis). To motivate variance models, note first that all the models that we studied in the class so far can be seen as "Mean Models".

For example, consider the high-dimensional linear regression model which can be written as:

$$y_t \overset{\text{ind}}{\sim} N(\mu_t, \sigma^2)$$

where ind stands for "independently distributed as". Note that the right hand side depends on t so the distribution of y_t changes with t and we cannot therefore use "i.i.d".

The parameters in this model are $\mu_1, \ldots, \mu_n$ and σ^2. Clearly this is a high-dimensional because the number of parameters is large.

If we attempt to estimate the parameters by maximizing the likelihood without any regularization, we get $\mu_t = y_t$ and $\sigma^2 = 0$, leading to full interpolation (overfitting) to the data. Regularization is therefore necessary to obtain something useful. If we want to obtain "smooth" trend estimates, we can employ regularization terms which force neighboring values or neighboring slopes of μ_t to be close. If we focus on slopes (which leads to more smoothness compared to just imposing closeness of values), we obtain the estimators $\hat{\mu}_t^{\text{ridge}}(\lambda)$ and $\hat{\mu}_t^{\text{lasso}}(\lambda)$ which minimize:

$$\sum_{t=1}^{n}(y_t - \mu_t)^2 + \lambda \sum_{t=2}^{n-1} \left((\mu_{t+1} - \mu_t) - (\mu_t - \mu_{t-1})\right)^2$$

and

$$\sum_{t=1}^{n}(y_t - \mu_t)^2 + \lambda \sum_{t=2}^{n-1} \left|(\mu_{t+1} - \mu_t) - (\mu_t - \mu_{t-1})\right|$$

respectively. We have already studied these estimators last week where we observed, among other things, that they can be alternatively represented as $\hat{\mu}_t^{\text{ridge}}(\lambda) = X\hat{\beta}_{\text{ridge}}(\lambda)$ and $\hat{\mu}_t^{\text{lasso}}(\lambda) = X\hat{\beta}_{\text{lasso}}(\lambda)$

where

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \text{ and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix} \tag{1}$$

and $\hat{\beta}_{\text{ridge}}(\lambda)$ and $\hat{\beta}_{\text{lasso}}(\lambda)$ minimize

$$\|y - X\beta\|^2 + \sum_{t=2}^{n-1} \beta_t^2$$

and

$$\|y - X\beta\|^2 + \sum_{t=2}^{n-1} |\beta_t|$$

respectively.

This model is an example of a "Mean Model" where the focus is on estimating the mean parameters μ_t. In contrast, we shall now study "Variance Models" which model the variances of the data. The simplest variance model is given by:

$$y_t \overset{\text{ind}}{\sim} N(0, \tau_t^2) \tag{2}$$

The parameters are $\tau_1^2, \ldots, \tau_n^2$. Since these represent variances, we refer to this as a "variance model". We shall study parameter estimation in this model (and other related models).

2 Three High-Dimensional Models

We shall discuss three simple high-dimensional models for time series $\{y_t\}$. The first model is the same as the high-dimensional linear regression model considered earlier; but we shall write it slightly differently. The second model is a model on variances. The third model is a variance model on the DFT and will be referred to as the Spectrum model.

2.1 Model One

This is the model

$$y_t \overset{\text{ind}}{\sim} N(\mu_t, \sigma^2)$$

where ind stands for "independently distributed as". Note that the right hand side depends on t so the distribution of y_t changes with t and we cannot therefore use "i.i.d".

The parameters in this model are $\mu_1, \ldots, \mu_n$ and σ^2. Clearly this is a high-dimensional because the number of parameters is large.

If we attempt to estimate the parameters by maximizing the likelihood without any regularization, we get $\mu_t = y_t$ and $\sigma^2 = 0$, leading to full interpolation (overfitting) to the data. Regularization is therefore necessary to obtain something useful. If we want to obtain "smooth" trend estimates, we can employ regularization terms which force neighboring values or neighboring slopes of μ_t to be

close. If we focus on slopes (which leads to more smoothness compared to just imposing closeness of values), we obtain the estimators $\hat{\mu}_t^{\text{ridge}}(\lambda)$ and $\hat{\mu}_t^{\text{lasso}}(\lambda)$ which minimize:

$$\sum_{t=1}^{n}(y_t - \mu_t)^2 + \lambda \sum_{t=2}^{n-1}\left((\mu_{t+1} - \mu_t) - (\mu_t - \mu_{t-1})\right)^2$$

and

$$\sum_{t=1}^{n}(y_t - \mu_t)^2 + \lambda \sum_{t=2}^{n-1}\left|(\mu_{t+1} - \mu_t) - (\mu_t - \mu_{t-1})\right|$$

respectively. We have already studied these estimators last week where we observed, among other things, that they can be alternatively represented as $\hat{\mu}_t^{\text{ridge}}(\lambda) = X\hat{\beta}_{\text{ridge}}(\lambda)$ and $\hat{\mu}_t^{\text{lasso}}(\lambda) = X\hat{\beta}_{\text{lasso}}(\lambda)$ where

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \text{ and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix} \tag{3}$$

and $\hat{\beta}_{\text{ridge}}(\lambda)$ and $\hat{\beta}_{\text{lasso}}(\lambda)$ minimize

$$\|y - X\beta\|^2 + \sum_{t=2}^{n-1}\beta_t^2$$

and

$$\|y - X\beta\|^2 + \sum_{t=2}^{n-1}|\beta_t|$$

respectively.

This model is an example of a "Mean Model" where the focus is on estimating the mean parameters μ_t. In contrast, the next two models will be examples of "Variance Models".

2.2 Model Two

This model is given by:

$$y_t \overset{\text{ind}}{\sim} N(0, \tau_t^2) \tag{4}$$

The parameters are $\tau_1^2, \ldots, \tau_n^2$. Since these represent variances, we refer to this as a "variance model". τ_t can be interpreted as the magnitude of y_t. This model is useful when we care only about the magnitudes of the observations y_t (and not their signs).

Model (4) (similar to Model One from the previous section) is also a high-dimensional model because the number of parameters is the same as the number of observations.

The likelihood for this model is proportional to

$$\prod_{t=1}^{n}\frac{1}{\tau_t}\exp\left(-\frac{y_t^2}{2\tau_t^2}\right). \tag{5}$$

Note that the likelihood depends on the data only through the squares $y_1^2, \ldots, y_n^2$. Therefore the squares $y_1^2, \ldots, y_n^2$ form the "sufficient statistic" in this model. Under (4), we have

$$y_t^2 \overset{\text{ind}}{\sim} \tau_t^2 \chi_1^2$$

where χ_1^2 denotes the chi-squared distribution with 1 degree of freedom. Instead of writing the likelihood in terms of the raw data y_t, we can also write the likelihood using the squares y_t^2. This will lead to a slightly different form for the likelihood that should still be proportional to (5). To see this, observe that the density of χ_1^2 is proportional to $x^{-1/2}\exp(-x/2)$ so that the density of $\tau_t^2\chi_1^2$ is proportional to

$$\frac{1}{\tau_t^2}\left(\frac{x}{\tau_t^2}\right)^{-1/2}\exp\left(-\frac{x}{2\tau_t^2}\right) = x^{-1/2}\frac{1}{\tau_t}\exp\left(-\frac{x}{2\tau_t^2}\right).$$

The likelihood written in terms of $y_1^2, \ldots, y_n^2$ is thus

$$\prod_{t=1}^{n}(y_t^2)^{-1/2}\frac{1}{\tau_t}\exp\left(-\frac{y_t^2}{2\tau_t^2}\right).$$

The term $(y_t^2)^{-1/2}$ above can be dropped as it is a constant not depending on the parameters τ_t^2. Dropping it leads to (5).

The log-likelihood is:

$$\sum_{t=1}^{n}\left(-\log\tau_t - \frac{y_t^2}{2\tau_t^2}\right).$$

It is a convention to write optimization problems for computing estimators as minimization problems (as opposed to maximization). For this, we write the negative log-likelihood which is given by:

$$\sum_{t=1}^{n}\left(\log\tau_t + \frac{y_t^2}{2\tau_t^2}\right).$$

As τ_t is a standard deviation parameter that is constrained to be positive, it is better to deal with $\alpha_t = \log\tau_t$ instead of τ_t directly. This reparameterization has the following benefits:

- **Unconstrained Optimization**: Unlike τ_t, which must be positive, α_t can take any real value, allowing for more stable numerical optimization.
- **Improved Computational Stability**: Variance parameters can vary over several orders of magnitude, and working in the log scale reduces numerical precision issues.

Because of these benefits, many variance modeling approaches use log-variance transformations (see e.g., stochastic volatility models).

Writing the negative log-likelihood in terms of $\alpha_t = \log\tau_t$, we get

$$\sum_{t=1}^{n}\left(\alpha_t + \frac{y_t^2}{2}e^{-2\alpha_t}\right)$$

If we minimize the above (without any additional regularization) with respect to α_t, we obtain $\alpha_t = \log|y_t|$, or equivalently, $\tau_t^2 = y_t^2$. In other words, the parameters τ_t^2 will fully interpolate (overfit) the sufficient statistics y_t^2.

For a more useful estimation procedure, we need to introduce regularization. If we assume that α_t is smooth, we can add the penalty $\sum_{t=2}^{n-1}((\alpha_{t+1}-\alpha_t)-(\alpha_t-\alpha_{t-1}))^2$ or $\sum_{t=2}^{n-1}|(\alpha_{t+1}-\alpha_t)-(\alpha_t-\alpha_{t-1})|$ to the negative log-likelihood. This leads to the estimators $\hat{\alpha}_t^{\text{ridge}}(\lambda)$ and $\hat{\alpha}_t^{\text{lasso}}(\lambda)$ which are defined as the minimizers of

$$\sum_{t=1}^{n}\left(\alpha_t+\frac{y_t^2}{2}e^{-2\alpha_t}\right)+\lambda\sum_{t=2}^{n-1}((\alpha_{t+1}-\alpha_t)-(\alpha_t-\alpha_{t-1}))^2$$

and

$$\sum_{t=1}^{n}\left(\alpha_t+\frac{y_t^2}{2}e^{-2\alpha_t}\right)+\lambda\sum_{t=2}^{n-1}|(\alpha_{t+1}-\alpha_t)-(\alpha_t-\alpha_{t-1})|$$

respectively. The penalties encourage smoothness in $\{\alpha_t\}$, leading to more stable and interpretable variance estimates. These optimizations are convex and they can be solved, for example, using functions from the python library `cvxpy` just as we solved $\hat{\beta}_{\text{ridge}}(\lambda)$ and $\hat{\beta}_{\text{lasso}}(\lambda)$ from the previous section.

2.3 Model Three

Model three is essentially Model two but applied to the DFT. In order to describe this, let us first revisit the DFT. Given a time series $y_0,\ldots,y_{n-1}$, its DFT is $b_0,b_1,\ldots,b_{n-1}$ where

$$b_j=\sum_{t=0}^{n-1}y_t\exp\left(-\frac{2\pi ijt}{n}\right).$$

b_j is a complex number with real and imaginary parts given by

$$\text{Re}(b_j)=\sum_{t=0}^{n-1}y_t\cos\left(\frac{2\pi jt}{n}\right)\quad\text{and}\quad\text{Im}(b_j)=-\sum_{t=0}^{n-1}y_t\sin\left(\frac{2\pi jt}{n}\right)$$

When $j=0$, the imaginary part is zero and we get $b_0=\sum_{t=0}^{n-1}y_t$. b_0 is therefore just the sum of the datapoints and it does not provide any information on cycles etc.

Another fact that we previously verified is $b_{n-j}=\bar{b}_j$ (here $\bar{b}_j$ denotes the complex conjugate of b_j). Because of this property, the later half of the DFT terms is redundant (as they can be recovered from the first half).

If n is odd and $m=(n-1)/2$, then the most important DFT terms are $b_1,\ldots,b_m$. The other terms are b_0 which is simply the sum of the data points and $b_{m+1},\ldots,b_{n-1}$ which are simply the complex conjugates of $b_m,b_{m-1},\ldots,b_1$.

If n is even and $m=(n-2)/2$, then the most important DFT terms are $b_1,\ldots,b_m$ and b_{m+1}. The other DFT terms are b_0 which is simply the sum of the data points and $b_{m+2},\ldots,b_{n-1}$ which are the complex conjugates of $b_m,\ldots,b_1$. Note in this case that $_{m+1}=b_{n/2}$ will have zero imaginary part (hence b_{m+1} is real).

Below we focus on the case where n is odd for simplicity, and take $m=(n-1)/2$. Model three is obtained by using Model two for the DFT terms $b_1,\ldots,b_m$. Because b_j can be complex, we use the modeling assumption for both the real and imaginary parts:

$$\text{Re}(b_j),\text{Im}(b_j)\overset{\text{i.i.d}}{\sim}N(0,\gamma_j^2)\qquad\text{for } j=1,\ldots,m. \tag{6}$$

We also assume that b_j are independent across j. The unknown parameters in this model are $\gamma_1, \ldots, \gamma_m$. γ_j represents the strength of the sinusoids at frequency j/n.

The likelihood corresponding to (6) is proportional to:

$$\prod_{j=1}^{m} \frac{1}{\gamma_j} \exp\left(-\frac{(\text{Re}(b_j))^2}{2\gamma_j^2}\right) \frac{1}{\gamma_j} \exp\left(-\frac{(\text{Im}(b_j))^2}{2\gamma_j^2}\right)$$
$$= \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{(\text{Re}(b_j))^2 + (\text{Im}(b_j))^2}{2\gamma_j^2}\right) = \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{|b_j|^2}{2\gamma_j^2}\right).$$

Therefore the likelihood depends on the squared magnitudes $|b_j|^2$ of the DFT coefficients. Recall that the periodogram $I(j/n)$ is defined as

$$I(j/n) := \frac{|b_j|^2}{n}.$$

We can therefore rewrite the likelihood in terms of the periodogram as follows:

$$\prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{nI(j/n)}{2\gamma_j^2}\right). \tag{7}$$

This likelihood depends on the data only through the periodogram ordinates $I(j/n)$ for $j = 1, \ldots, m$. Therefore the periodogram forms the sufficient statistic in this model. Under (6), we have

$$I(j/n) = \frac{1}{n}|b_j|^2 = \frac{1}{n}\left((\text{Re}(b_j))^2 + (\text{Im}(b_j))^2\right) \sim \frac{\gamma_j^2}{n}\chi_2^2.$$

The model can therefore be written directly in terms of the periodogram as

$$I(j/n) \overset{\text{ind}}{\sim} \frac{\gamma_j^2}{n}\chi_2^2 \qquad \text{for } j = 1, \ldots, m.$$

We can write the likelihood for the above model in terms of the periodogram and this would be proportional to (7). Note also that χ_2^2 distribution with two degrees of freedom actually coincides with the Exponential distribution with λ parameter equal to $1/2$.

Intuitively, Model (6) does not care so much about the individual DFT coefficients b_j but only their magnitude.

The negative log-likelihood corresponding to (7) is

$$\sum_{j=1}^{m} \left(2\log\gamma_j + \frac{nI(j/n)}{2\gamma_j^2}\right).$$

As in the case of Model two, for optimization purposes we work with the logarithms of γ_j. Let $\alpha_j = \log\gamma_j$. The negative log-likelihood in terms of α_j is

$$\sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j}\right).$$

If we directly minimize the above with respect to α_j (without any additional regularization), we get

$$\alpha_j = \log\sqrt{\frac{nI(j/n)}{2}} \quad \text{and} \quad \gamma_j^2 = e^{2\alpha_j} = \frac{nI(j/n)}{2}.$$

This basically means that the γ_j^2 parameters fully interpolate the periodogram leading to full overfitting. For more meaningful estimation, we need to add regularization. If we assume smoothness of α_j, we can add the penalty $\sum_{j=2}^{m-1}((\alpha_{j+1}-\alpha_j)-(\alpha_j-\alpha_{j-1}))^2$ or $\sum_{j=2}^{m-1}|(\alpha_{j+1}-\alpha_j)-(\alpha_j-\alpha_{j-1})|$ to the negative log-likelihood. This leads to the estimators $\hat{\alpha}_t^{\text{ridge}}(\lambda)$ and $\hat{\alpha}_t^{\text{lasso}}(\lambda)$ which are defined as the minimizers of

$$\sum_{j=1}^{m}\left(2\alpha_j+\frac{nI(j/n)}{2}e^{-2\alpha_j}\right)+\lambda\sum_{j=2}^{m-1}\left((\alpha_{j+1}-\alpha_j)-(\alpha_j-\alpha_{j-1})\right)^2$$

and

$$\sum_{t=1}^{n}\left(2\alpha_j+\frac{nI(j/n)}{2}e^{-2\alpha_j}\right)+\lambda\sum_{j=2}^{m-1}|(\alpha_{j+1}-\alpha_j)-(\alpha_j-\alpha_{j-1})|$$

respectively. The penalties encourage smoothness in $\{\alpha_j\}$, leading to more stable and interpretable estimates for $\{\gamma_j\}$.

3 Spectral Analysis

3.1 Smoothing the Periodogram

Consider a time series dataset $y_0,\ldots,y_{n-1}$. One of most common tools in time data analysis is the periodogram. It is given by:

$$I(j/n):=\frac{|b_j|^2}{n}\qquad\text{for } 0<\frac{j}{n}<1/2$$

where b_j is the Discrete Fourier Transform (DFT) defined by

$$b_j:=\sum_{t=0}^{n-1}y_t\exp\left(-\frac{2\pi ijt}{n}\right).$$

Let us assume that n is odd and $m=(n-1)/2$. Then the periodogram ordinate is defined for $j=1,\ldots,m$.

The periodogram shows which frequency components contribute most strongly to the behaviour of the observed time series. It is useful for identifying the dominant cyclical patterns or periodicities within a dataset.

For most real datasets, the periodogram appears rough and noisy, especially when looking at its plot on the log-scale (i.e., when looking at the plot of log-periodogram). In such cases, researchers attempt to smooth the periodogram to get a clean summary of the trend in the periodogram without getting distracted by noise. What is a good way of smoothing the periodogram?

A nice way of smoothing is via the use of an appropriate model. Previously, we saw for estimating a smooth trend function, say μ_t, from observed time series data y_t, a natural model is:

$$y_t=\mu_t+\epsilon_t\qquad\text{with } \epsilon_t\overset{\text{i.i.d}}{\sim}N(0,\sigma^2).$$

We can try to use a similar model for the periodogram. Modeling $I(j/n)$ as a smooth trend function plus Gaussian noise does not make much sense for multiple reasons: (a) the periodogram is a

positive quantity so using the normal distribution may not be a good idea, (b) the noise in the periodogram is more pronounced on the log-scale so we should probably use an additive noise model for the log-periodogram as opposed to the periodogram directly. In other words, we should use a multiplicative noise model for the periodogram.

The model that we shall use here is:

$$I(j/n) = f(j/n)\eta_j \qquad \text{for } j = 1, \ldots, m.$$

where $f(j/n)$ is a smooth function that represents the trend in the periodogram, and

$$\eta_j \overset{\text{i.i.d}}{\sim} \frac{1}{2}\chi^2_2.$$

χ^2_2 is the chi-squared distribution with 2 degrees of freedom; its density is given by:

$$f_{\chi^2_2}(x) := \frac{1}{2}\exp(-x/2)I\{x > 0\}.$$

The density of $\chi^2_2/2$ is then given by:

$$f_{\chi^2_2/2}(x) = 2f_{\chi^2_2}(2x) = \exp(-x)I\{x > 0\}.$$

The above is the density of the Standard Exponential Distribution (the exponential distribution $Exp(\lambda)$ has density $\lambda e^{-\lambda x}I\{x > 0\}$; the above corresponds to $\lambda = 1$). Thus

$$\frac{\chi^2_2}{2} = Exp(1).$$

Our model can therefore also be written as:

$$I(j/n) = f(j/n)\eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} Exp(1). \tag{8}$$

This is a multiplicative noise model. We can rewrite it in additive form by taking logarithms:

$$\log I(j/n) = \log f(j/n) + \log \eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} Exp(1). \tag{9}$$

The noise in this additive representation is captured by the terms $\log \eta_j, 1 \le j \le n$. These variables do not have mean zero (from the internet $\mathbb{E}\log\eta_j \approx -0.5772$). Therefore $\log f(j/n)$ does the represent the mean of $\log I(j/n)$ (in fact, it is strictly larger than the mean of $\log I(j/n)$). Rather, $f(j/n)$ represents the mean of $I(j/n)$.

We shall refer to (8) or (9) as the Spectrum Model.

3.2 Power Spectral Density

Since the mean of the $Exp(1)$ distribution equals 1, the mean of $I(j/n)$ in the Spectrum Model (8) is given by $f(j/n)$. This quantity is known as the **power** of frequency j/n:

$$f(j/n) := \text{power of frequency } j/n = \text{mean of } I(j/n) \text{ in model (8)}.$$

If we plot the points $(j/n, f(j/n))$ for $j = 1, \ldots, m$ and join the neighboring points by lines, we get a continuous function plot. This function is known as the **power spectral density** and is defined on $[0, 0.5]$.

This definition of the power spectral density is not rigorous. For a rigorous treatment, see any book on time series (e.g., Chapter 4 of Shumway and Stoffer; or the book "Spectral Analysis for Univariate Time Series" by Percival and Walden).

Note that the power spectral density is not a probability density function in the sense that it does not integrate to one.

3.3 Spectrum Model from DFT

In (8) or (9), we formulated the spectrum model in terms of the periodogram or log-periodogram. We can also describe it in terms of the DFT. Specifically, in terms of the DFT $b_0, \ldots, b_{n-1}$ of the data $y_0, \ldots, y_{n-1}$, the model is given by:

$$\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{\text{i.i.d}}{\sim} N(0, \gamma_j^2) \tag{10}$$

for $j = 1, \ldots, m$ where $m = (n-1)/2$ (we are assuming that n is odd). The unknown parameters in this model are $\gamma_1^2, \ldots, \gamma_m^2$ and γ_j represents the strength of sinusoids at frequency j/n.

By definition of the periodogram, we get

$$I(j/n) := \frac{|b_j|^2}{n} = \frac{1}{n}\left(\mathrm{Re}(b_j)^2 + \mathrm{Im}(b_j)^2\right) = \frac{2\gamma_j^2}{n}\frac{1}{2}\left(\left(\frac{\mathrm{Re}(b_j)}{\gamma_j}\right)^2 + \left(\frac{\mathrm{Im}(b_j)}{\gamma_j}\right)^2\right).$$

By the assumption (10), we get

$$\eta_j := \left(\left(\frac{\mathrm{Re}(b_j)}{\gamma_j}\right)^2 + \left(\frac{\mathrm{Im}(b_j)}{\gamma_j}\right)^2\right) \overset{\text{i.i.d}}{\sim} \chi_2^2.$$

Therefore the model (10) implies that

$$I(j/n) = \frac{2\gamma_j^2}{n}\eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} \frac{\chi_2^2}{2} = Exp(1).$$

The is exactly the same as (8) with the identification:

$$f(j/n) = \frac{2\gamma_j^2}{n}.$$

Because of this equivalence, we shall treat (10) as another formulation of the spectrum model.

3.4 Rewriting the Model in terms of y_t

The model (10) is written in terms of the DFT b_j. Because the original data can be written in terms of the DFT, we can convert (10) into a specification for the original data y_t. This will lead us to another model representation which is directly in terms of the original data y_t.

The key to this is the following formula which writes the data in terms of the DFT.

$$y_t = \frac{1}{n}\sum_{j=0}^{n-1} b_j \exp\left(\frac{2\pi i j t}{n}\right) \qquad \text{for } t = 0, 1, \ldots, n-1. \tag{11}$$

This is known as the **Inverse DFT** formula.

The right hand side of (11) involves complex numbers (b_j and $\exp(2\pi ijt/n)$). On the other hand, the left hand is the data y_t which is always real. Below we change the right hand side in (11) to make it consist of only real terms.

We can rewrite the inverse DFT formula in the following way.

$$\begin{aligned}
y_t &= \frac{1}{n}\sum_{j=0}^{n-1} b_j \exp\left(\frac{2\pi i j t}{n}\right) \\
&= \frac{1}{n}\sum_{j=0}^{n-1} (\mathrm{Re}(b_j) + i\ \mathrm{Im}(b_j))\left(\cos\left(\frac{2\pi j t}{n}\right) + i\sin\left(\frac{2\pi j t}{n}\right)\right) \\
&= \frac{1}{n}\sum_{j=0}^{n-1}\left(\mathrm{Re}(b_j)\cos\left(\frac{2\pi j t}{n}\right) - \mathrm{Im}(b_j)\sin\left(\frac{2\pi j t}{n}\right)\right) + i\frac{1}{n}\sum_{j=0}^{n-1}\left(\mathrm{Re}(b_j)\sin\left(\frac{2\pi j t}{n}\right) + \mathrm{Im}(b_j)\cos\left(\frac{2\pi j t}{n}\right)\right)
\end{aligned}$$

We can ignore the imaginary part above as the dataset consists of real numbers, and this leads to

$$\begin{aligned}
y_t &= \frac{1}{n}\sum_{j=0}^{n-1}\left(\mathrm{Re}(b_j)\cos\left(\frac{2\pi j t}{n}\right) - \mathrm{Im}(b_j)\sin\left(\frac{2\pi j t}{n}\right)\right) \\
&= \frac{b_0}{n} + \frac{1}{n}\sum_{j=1}^{n-1}\left(\mathrm{Re}(b_j)\cos\left(\frac{2\pi j t}{n}\right) - \mathrm{Im}(b_j)\sin\left(\frac{2\pi j t}{n}\right)\right)
\end{aligned}$$

We are assuming that n is odd and that $m = (n-1)/2$. We split the sum above into $j = 1, \ldots, m$ and then $j = m+1, \ldots, n-1$, and then use $b_{n-j} = \bar{b}_j$ or, equivalently, $\mathrm{Re}(b_{n-j}) = \mathrm{Re}(b_j)$ and $\mathrm{Im}(b_{n-j}) = -\mathrm{Im}(b_j)$. This gives

$$y_t = \frac{b_0}{n} + \sum_{j=1}^{m}\left(\frac{2\,\mathrm{Re}(b_j)}{n}\cos\left(\frac{2\pi j t}{n}\right) + \frac{-2\,\mathrm{Im}(b_j)}{n}\sin\left(\frac{2\pi j t}{n}\right)\right).$$

where we also used $\cos(2\pi(n-j)t/n) = \cos(2\pi j t/n)$ and $\sin(2\pi(n-j)t/n) = -\sin(2\pi j t/n)$.

In other words, when n is odd and $m = (n-1)/2$, we have

$$y_t = \beta_0 + \sum_{j=1}^{m}\left(\beta_{1j}\cos\frac{2\pi j t}{n} + \beta_{2j}\sin\frac{2\pi j t}{n}\right) \tag{12}$$

where, for $j = 1, \ldots, m$,

$$\beta_0 = \frac{b_0}{n} \quad \beta_{1j} = \frac{2\,\mathrm{Re}(b_j)}{n} \quad \beta_{2j} = -\frac{2\,\mathrm{Im}(b_j)}{n} \tag{13}$$

The formula (12) holds for every dataset $y_0, \ldots, y_{n-1}$. As a result, the model (10) is equivalent to (12) with

$$\beta_{1j} = \frac{2\,\mathrm{Re}(b_j)}{n} \sim N\left(0, \frac{4}{n^2}\gamma_j^2\right) \quad \text{and} \quad \beta_{2j} = \frac{-2\,\mathrm{Im}(b_j)}{n} \sim N\left(0, \frac{4}{n^2}\gamma_j^2\right).$$